

# Enterprise Linux Server Maintenance & AWS Support Project

**Client:** TechNova Solutions Pvt. Ltd.

**Role:** Linux Technical Support Engineer (L1 Support)

**Duration:** Jan 2023 – Present

**Environment:** Linux (RHEL/Ubuntu), AWS (EC2, EBS, CloudWatch), Shell Scripting

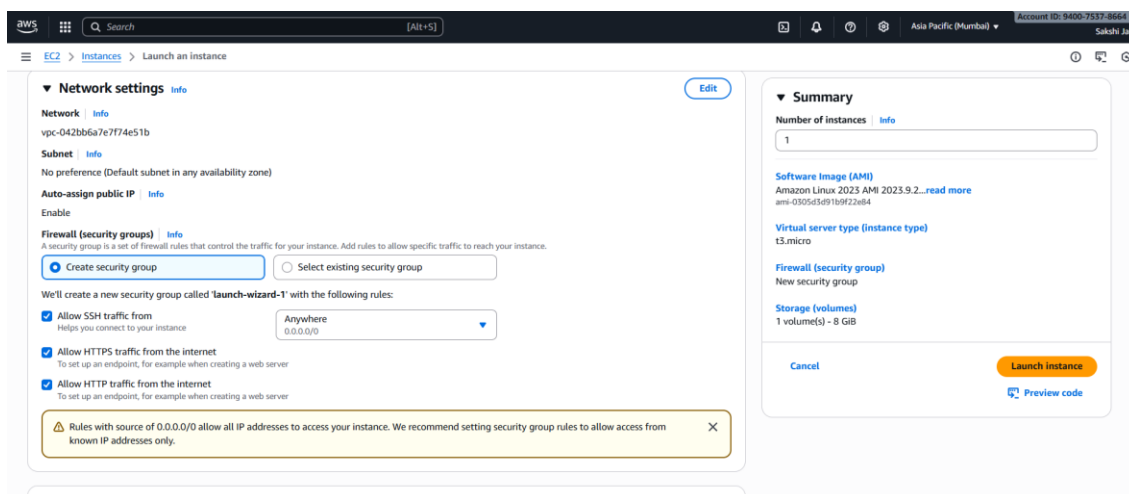
## 1. Objective

Maintain and support Linux-based servers (RHEL/Ubuntu) and AWS EC2 instances. Automate maintenance tasks, monitor system health, and ensure high uptime.

## 2. Server Provisioning and Setup

### Step 1: Launch EC2 Instance

- Login to AWS Console → EC2 → Launch Instance
- Choose:
  - **AMI:** Amazon Linux 2 / Ubuntu 20.04 / RHEL 8
  - **Instance Type:** t3.micro
  - **Storage:** Add EBS volume (20–30 GB)
  - **Security Group:** Allow SSH (22), HTTP (80), HTTPS (443)



## Step 2: Connect to Instance via SSH

**Connect** Info

Connect to an instance using the browser-based client.

**EC2 Instance Connect** | Session Manager | SSH client | EC2 serial console

**Instance ID**  
i-0a0f63691800c2f86 (TechNova-Instance)

**Connection type**

☒ Connect using a Public IP  
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP  
Connect using a private IP address and a VPC endpoint

**Public IPv4 address**  
13.203.155.8

**IPv6 address**  
--

**Username**  
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.  
ec2-user

**Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#) [Connect](#)

## Step 3: Basic System Configuration

```
yum update -y # RHEL / Amazon Linux
```

## Step 4: Create a New User & Grant Access

```
adduser adminuser
```

```
passwd adminuser
```

```
usermod -aG wheel adminuser
```

```
[root@webserver1 ~]# bash
[root@webserver1 ~]# adduser adminuser
[root@webserver1 ~]# passwd adminuser
Changing password for user adminuser.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[root@webserver1 ~]# usermod -aG wheel adminuser
```

```
[root@webserver1 ~]# tail /etc/passwd
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
adminuser:x:1001:1001::/home/adminuser:/bin/bash
```

## Step 5: Enable Firewall and Secure SSH

yum install firewalld -y

systemctl enable firewalld

firewall-cmd --add-service=ssh --permanent

firewall-cmd --add-service=http --permanent

firewall-cmd --reload

```
[root@webserver1 ~]# yum install firewalld -y
Last metadata expiration check: 0:28:19 ago on Sun Nov  9 10:35:52 2025.
Dependencies resolved.
=====
Package                               Architecture Version                               Repository
Size
-----
Installing:
firewalld                             noarch      1.2.3-1.amzn2023                     amazonlinux
452 k
Installing dependencies:
firewalld-filesystem                  noarch      1.2.3-1.amzn2023                     amazonlinux
11 k
gobject-introspection                 x86_64      1.82.0-1.amzn2023                   amazonlinux
122 k
ipset                                  x86_64      7.11-1.amzn2023.0.3                 amazonlinux
40 k
ipset-libs                            x86_64      7.11-1.amzn2023.0.3                 amazonlinux
67 k
iptables-libs                         x86_64      1.8.8-3.amzn2023.0.2                amazonlinux
401 k
iptables-nft                          x86_64      1.8.8-3.amzn2023.0.2                amazonlinux
183 k
libnetfilter_conntrack                x86_64      1.0.8-2.amzn2023.0.2                amazonlinux
58 k
libnftnl                              x86_64      1.0.1-19.amzn2023.0.2               amazonlinux
30 k
libnftnl                              x86_64      1.2.2-2.amzn2023.0.2                amazonlinux
84 k
nftables                              x86_64      1:1.0.4-3.amzn2023.0.2              amazonlinux
400 k
python3-firewall                      noarch      1.2.3-1.amzn2023                     amazonlinux

[root@webserver1 ~]# systemctl enable firewalld
[root@webserver1 ~]# systemctl start firewalld
[root@webserver1 ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-11-09 11:06:00 IST; 6s ago
     Docs: man:firewalld(1)
    Main PID: 26748 (firewalld)
      Tasks: 2 (limit: 1053)
    Memory: 24.5M
      CPU: 410ms
    CGroup: /system.slice/firewalld.service
            └─26748 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Nov 09 11:06:00 webserver1 systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
Nov 09 11:06:00 webserver1 systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
```

```

[root@webserver1 ~]# firewall-cmd --add-service=ssh --permanent
Warning: ALREADY_ENABLED: ssh
success
[root@webserver1 ~]# firewall-cmd --add-service=http --permanent
success
[root@webserver1 ~]# firewall-cmd --reload
success
[root@webserver1 ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client http mdns ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@webserver1 ~]#

```

# Disable root SSH login

vim /etc/ssh/sshd\_config

# Change: PermitRootLogin no

systemctl restart sshd

```

25
26 #HostKey /etc/ssh/ssh_host_rsa_key
27 #HostKey /etc/ssh/ssh_host_ecdsa_key
28 #HostKey /etc/ssh/ssh_host_ed25519_key
29
30 # Ciphers and keying
31 #RekeyLimit default none
32
33 # Logging
34 #SyslogFacility AUTH
35 #LogLevel INFO
36
37 # Authentication:
38
39 #LoginGraceTime 2m
40 PermitRootLogin No
41 # prohibit-password
42 #StrictModes yes
43 #MaxAuthTries 6
44 #MaxSessions 10
45
46 #PubkeyAuthentication yes
47
48 # The default is to check both .ssh/authorized keys and .ssh/authorized_keys2
49 # but this is overridden so installations will only check .ssh/authorized_keys
50 AuthorizedKeysFile .ssh/authorized keys
-- INSERT --

```

40,19 21%

```

[root@webserver1 ~]# vim /etc/ssh/sshd_config
[root@webserver1 ~]# systemctl restart sshd
[root@webserver1 ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-11-09 11:21:45 IST; 34s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 27702 (sshd)
    Tasks: 1 (limit: 1053)
   Memory: 1.3M
      CPU: 9ms
  CGroup: /system.slice/ssh.service
          └─27702 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 09 11:21:45 webserver1 systemd[1]: Starting sshd.service - OpenSSH server daemon...
Nov 09 11:21:45 webserver1 sshd[27702]: Server listening on 0.0.0.0 port 22.
Nov 09 11:21:45 webserver1 sshd[27702]: Server listening on :: port 22.
Nov 09 11:21:45 webserver1 systemd[1]: Started sshd.service - OpenSSH server daemon.
[root@webserver1 ~]# █

```

### 3. Routine Maintenance & Automation

#### Step 6: Create a Shell Script for Disk Usage Check

/home/adminuser/scripts/disk\_usage.sh

```
#!/bin/bash
```

```
LOGFILE="/var/log/disk_report.log"
```

```
echo "Disk Usage Report - $(date)" >> $LOGFILE
```

```
df -h >> $LOGFILE
```

```
echo "-----" >> $LOGFILE
```

```

[root@webserver1 ~]# cd /home/adminuser/
[root@webserver1 adminuser]# ls
[root@webserver1 adminuser]# mkdir /scripts
[root@webserver1 adminuser]# ls
[root@webserver1 adminuser]# pwd
/home/adminuser
[root@webserver1 adminuser]# cd /scripts
[root@webserver1 scripts]# ls
[root@webserver1 scripts]# touch disk_usage.sh
[root@webserver1 scripts]# █

```

```
#!/bin/bash
LOGFILE="/var/log/disk_report.log"
echo "Disk Usage Report - $(date)" >> $LOGFILE
df -h >> $LOGFILE
echo "-----" >> $LOGFILE
~
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~
-- INSERT --
ll
```

## Step 7: Schedule via Cron

```
crontab -e
```

# Run daily at 11:50 AM

```
50 11 * * * /home/adminuser/scripts/disk_usage.sh
```

```
[root@webserver1 scripts]# yum install cronie -y
Last metadata expiration check: 0:57:51 ago on Sun Nov  9 10:35:52 2025.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository
ze
=====
Installing:
cronie                                x86_64            1.5.7-1.amzn2023.0.2  amazonlinux
k
Installing dependencies:
cronie-anacron                        x86_64            1.5.7-1.amzn2023.0.2  amazonlinux
k
Transaction Summary
=====
Install 2 Packages

Total download size: 147 k
Installed size: 341 k
Downloading Packages:
(1/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm           0% [=====] --
(1/2): cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64.rpm    1.
(2/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm          21% [=====] 1.
(2/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm          2.
```

```
50 11 * * * /home/adminuser/scripts/disk_usage.sh
~
~
```

```
[root@webserver1 /]# crontab -l
50 11 * * * /home/adminuser/scripts/disk_usage.sh
```

## Step 8: Create Log Cleanup Script

/home/adminuser/scripts/cleanup\_logs.sh

```
#!/bin/bash
```

```
find /var/log -type f -name "*.log" -mtime +30 -exec rm -f {} \;
```

```
echo "Old logs cleaned on $(date)" >> /var/log/cleanup.log
```

Schedule Weekly Cleanup:

```
crontab -e
```

```
# Every Sunday at 2 AM
```

```
0 2 * * 0 /home/adminuser/scripts/cleanup_logs.sh
```

```
#!/bin/bash
find /var/log -type f -name "*.log" -mtime +30 -exec rm -f {} \;
echo "Old logs cleaned on $(date)" >> /var/log/cleanup.log
#Schedule Weekly Cleanup:
crontab -e
# Every Sunday at 2 AM
0 2 * * 0 /home/adminuser/scripts/cleanup_logs.sh

-- INSERT --

11
```

## 4. AWS EBS Volume Management

## Step 9: Create and Attach New Volume

```
aws ec2 create-volume --size 20 --region ap-south-1 --availability-zone ap-south-1a --volume-type gp3
```

```
aws ec2 attach-volume --volume-id vol-0abcd1234 --instance-id i-0123456789abcdef0 --device /dev/sdf
```

EC2 > Volumes > Create volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3)

Size (GiB) [Info](#)

10

Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

aps1-az3 (ap-south-1b)

Snapshot ID - optional [Info](#)

aws

Search

[Alt++5]

2

🔔

🔄

⚙️

Asia Pacific (Mumbai)

Account ID: 9400-7537-0664

Sakshi J

☰

EC2 > Volumes > vol-084476d03c77c3dc3 > Attach volume

🔔

🖨️

🌐

## Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

### Basic details

Volume ID  
🔗 vol-084476d03c77c3dc3

Availability Zone  
aps1-az3 (ap-south-1b)

Instance Info  
i-0a0f63691800c2f86  
(TechNova-Instance) [running] 🔄

Only instances in the same Availability Zone as the selected volume are displayed.

Device name Info  
/dev/sdb

Recommended device names for Linux: /dev/xvda for root volume, /dev/sdf-pj for data volumes.

🔔 Newer Linux kernels may rename your devices to **/dev/xvdf** through **/dev/xvdp** internally, even when the device name entered here (and shown in the details) is **/dev/sdf** through **/dev/sdp**.

Cancel

Attach volume

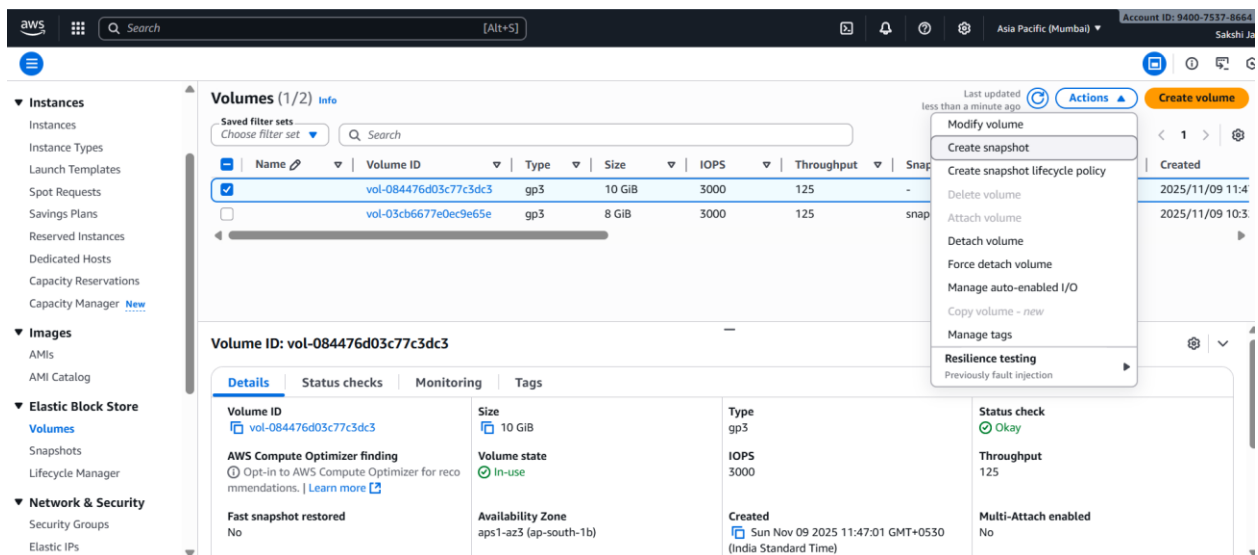
## Step 10: Format and Mount the Volume

```
mkfs.xfs /dev/sdb
```

```
mount /dev/sdb /data
```

```
[root@webserver1 ~]# lsblk
NAME                                MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1                            259:0    0  8G  0 disk
├─nvme0n1p1                        259:1    0  8G  0 part /
├─nvme0n1p127                     259:2    0  1M  0 part
└─nvme0n1p128                     259:3    0 10M  0 part /boot/efi
nvme1n1                            259:4    0 10G  0 disk
[root@webserver1 ~]# mkfs.xfs /dev/sdb
meta-data=/dev/sdb              isize=512    agcount=16, agsize=163840 blks
=                               sectsz=512   attr=2, projid32bit=1
=                               crc=1      finobt=1, sparse=1, rmapbt=0
=                               reflink=1   bigtime=1 inobtcount=1 nrext64=0
=                               exchange=0
data                =          bsize=4096   blocks=2621440, imaxpct=25
=                               sunit=1    swidth=1 blks
naming              =version 2          bsize=4096   ascii-ci=0, ftype=1, parent=0
log                 =internal log      bsize=4096   blocks=16384, version=2
=                               sectsz=512   sunit=1 blks, lazy-count=1
realtime            =none              extsz=4096   blocks=0, rtextents=0
[root@webserver1 ~]# mkfs.xfs /dev/sdb
mkfs.xfs: /dev/sdb appears to contain an existing filesystem (xfs).
mkfs.xfs: Use the -f option to force overwrite.
[root@webserver1 ~]# mkdir /data
[root@webserver1 ~]# mount /dev/sdb /data
[root@webserver1 ~]# lsblk
NAME                                MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1                            259:0    0  8G  0 disk
├─nvme0n1p1                        259:1    0  8G  0 part /
├─nvme0n1p127                     259:2    0  1M  0 part
└─nvme0n1p128                     259:3    0 10M  0 part /boot/efi
nvme1n1                            259:4    0 10G  0 disk /data
[root@webserver1 ~]# echo "/dev/sdb /data xfs defaults 0 0" | tee -a /etc/fstab
/dev/sdb /data xfs defaults 0 0
[root@webserver1 ~]#
```

## Step 11: Create EBS Snapshot Backup

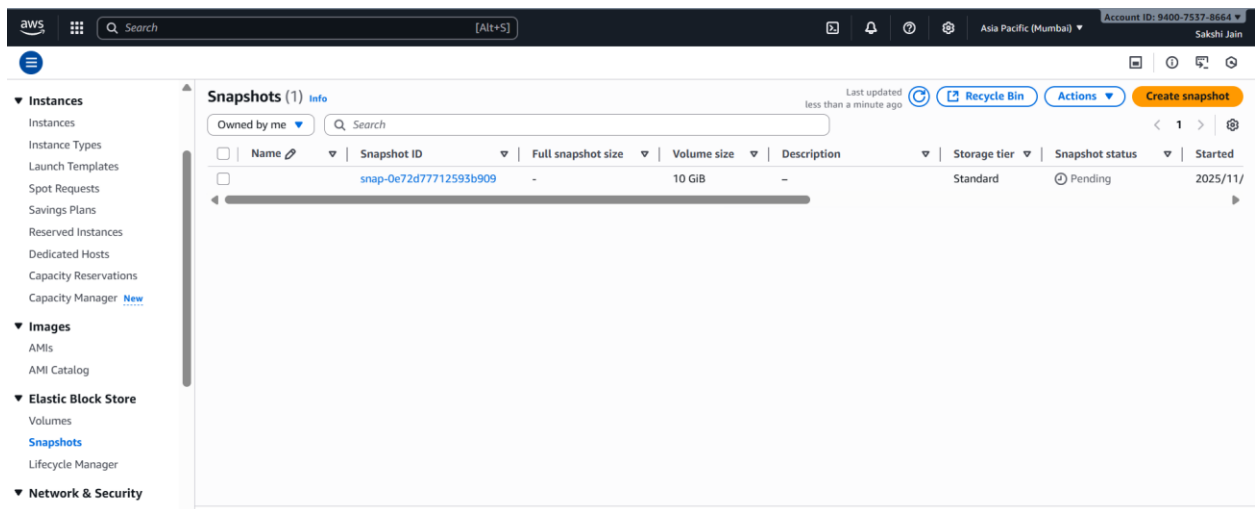


The screenshot shows the AWS Management Console interface for the 'Volumes' section. The left sidebar contains navigation links for Instances, Images, Elastic Block Store, and Network & Security. The main content area displays a table of EBS volumes. The first volume, 'vol-084476d03c77c3dc3', is selected. An 'Actions' dropdown menu is open, showing options like 'Modify volume', 'Create snapshot', 'Delete volume', etc. The 'Create snapshot' option is highlighted. Below the table, the details for the selected volume are shown, including its size (10 GiB), type (gp3), and status (In-use).

| Name                                | Volume ID             | Type | Size   | IOPS | Throughput | Snapshot |
|-------------------------------------|-----------------------|------|--------|------|------------|----------|
| <input checked="" type="checkbox"/> | vol-084476d03c77c3dc3 | gp3  | 10 GiB | 3000 | 125        | -        |
| <input type="checkbox"/>            | vol-03cb6677e0ec9e65e | gp3  | 8 GiB  | 3000 | 125        | snap-    |

Volume ID: vol-084476d03c77c3dc3

| Details                                                                                                                                                | Status checks                                          | Monitoring                                | Tags                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Volume ID</b><br>vol-084476d03c77c3dc3<br><b>AWS Compute Optimizer finding</b><br>Opt-in to AWS Compute Optimizer for recommendations.   Learn more | <b>Size</b><br>10 GiB<br><b>Volume state</b><br>In-use | <b>Type</b><br>gp3<br><b>IOPS</b><br>3000 | <b>Status check</b><br>Okay<br><b>Throughput</b><br>125<br><b>Multi-Attach enabled</b><br>No |



The screenshot shows the AWS Management Console interface for the 'Snapshots' section. The left sidebar contains navigation links for Instances, Images, Elastic Block Store, and Network & Security. The main content area displays a table of EBS snapshots. The first snapshot, 'snap-0e72d77712595b909', is shown with a status of 'Pending'. The 'Create snapshot' button is visible in the top right corner.

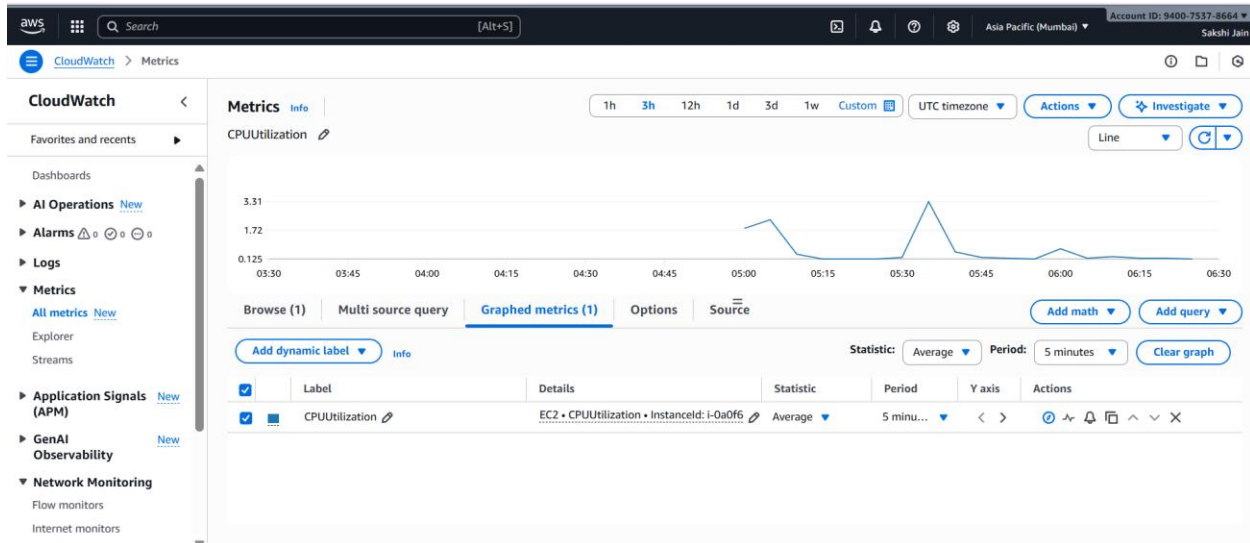
| Name                     | Snapshot ID            | Full snapshot size | Volume size | Description | Storage tier | Snapshot status | Started  |
|--------------------------|------------------------|--------------------|-------------|-------------|--------------|-----------------|----------|
| <input type="checkbox"/> | snap-0e72d77712595b909 | -                  | 10 GiB      | -           | Standard     | Pending         | 2025/11/ |

## 5. Monitoring and Alerts (CloudWatch + SNS)

### Step 12: Enable CloudWatch Metrics

Default EC2 metrics include CPU, Disk I/O, and Status Checks.

**CloudWatch → Metrics → EC2 → Per-Instance Metrics → pick your InstanceId and CPUUtilization**



### Step 13: Create CloudWatch Alarm

```
aws cloudwatch put-metric-alarm \
--alarm-name "High-CPU-Alarm" \
--metric-name CPUUtilization \
--namespace AWS/EC2 \
--statistic Average \
--period 300 \
--threshold 80 \
--comparison-operator GreaterThanThreshold \
--dimensions Name=InstanceId,Value=i-0123456789abcdef0 \
--evaluation-periods 2 \
--alarm-actions arn:aws:sns:ap-south-1:123456789012:OpsAlerts \
--treat-missing-data notBreaching
```

### Step 14: Create SNS Topic and Subscribe Email

```
aws sns create-topic --name OpsAlerts

aws sns subscribe --topic-arn arn:aws:sns:ap-south-1:123456789012:OpsAlerts --protocol email -
-notification-endpoint yourmail@example.com
```

## 6. System Monitoring Commands

### Daily Health Check

uptime

df -h

free -m

```
[root@webserver1 ~]# uptime
12:37:15 up 2:04, 2 users, load average: 0.00, 0.01, 0.00
[root@webserver1 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M   0    4.0M   0% /dev
tmpfs            453M   0    453M   0% /dev/shm
tmpfs           181M  444K   181M   1% /run
/dev/nvme0n1p1   8.0G  1.7G   6.4G  21% /
tmpfs            453M   0    453M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
/dev/nvme1n1     10G  104M   9.9G   2% /data
tmpfs            91M   0     91M   0% /run/user/1000
[root@webserver1 ~]# free -m
              total        used          free      shared  buff/cache   available
Mem:           904          221           298           0         384         544
Swap:            0            0            0
```

top -n 1 | head -15

```
top - 12:37:59 up 2:05, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.2 us, 0.0 sy, 0.0 ni, 96.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 904.8 total, 298.7 free, 221.7 used, 384.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 544.5 avail Mem
```

| PID   | USER | PR | NI  | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND                     |
|-------|------|----|-----|--------|-------|-------|---|------|------|---------|-----------------------------|
| 30757 | root | 20 | 0   | 224048 | 3332  | 2780  | R | 6.7  | 0.4  | 0:00.01 | top                         |
| 1     | root | 20 | 0   | 172880 | 17684 | 10760 | S | 0.0  | 1.9  | 0:01.67 | systemd                     |
| 2     | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | kthreadd                    |
| 3     | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_gp                      |
| 4     | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_par_gp                  |
| 5     | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | slub_flushwq                |
| 6     | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | netns                       |
| 8     | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | kworker/0:0H-events_highpri |

vmstat 1 5 # sample memory & CPU every second 5 times

iostat 1 5

```
[root@webserver1 ~]# vmstat 1 5
procs-----memory-----swap-----io-----system-----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st
 0 0 0 305844 2168 391516 0 0 16 40 26 45 0 0 100 0 0
 0 0 0 305844 2168 391516 0 0 0 0 42 65 0 0 100 0 0
 0 0 0 305844 2168 391516 0 0 0 0 43 72 0 0 100 0 0
 0 0 0 305844 2168 391516 0 0 0 0 38 57 0 0 100 0 0
 0 0 0 305844 2168 391516 0 0 0 0 35 61 0 0 100 0 0
[root@webserver1 ~]# iostat 1 5
Linux 6.1.156-177.286.amzn2023.x86_64 (webserver1) 11/09/25 _x86_64_ (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.26    0.02    0.14    0.05    0.01   99.52

Device:            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1             2.80         32.06         71.07         0.00     242403     537336         0
nvme1n1             0.12          0.74          9.04         0.00       5578      68383         0

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.00    0.00    0.00    0.00    0.00  100.00

Device:            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1             0.00         0.00         0.00         0.00         0         0         0
nvme1n1             0.00         0.00         0.00         0.00         0         0         0

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.00    0.00    0.00    0.00    0.00  100.00
```

sar 1 5

journalctl -xe

```
[root@webserver1 ~]# sar 1 5
Linux 6.1.156-177.286.amzn2023.x86_64 (webserver1) 11/09/25 _x86_64_ (2 CPU)

12:39:09      CPU      %user   %nice   %system %iowait  %steal   %idle
12:39:10      all       0.00    0.00    0.00    0.00    0.00   100.00
12:39:11      all       0.00    0.00    0.00    0.00    0.00   100.00
12:39:12      all       0.00    0.00    0.00    0.00    0.00   100.00
12:39:13      all       0.00    0.00    0.00    0.00    0.00   100.00
12:39:14      all       0.00    0.00    0.00    0.00    0.00   100.00
Average:      all       0.00    0.00    0.00    0.00    0.00   100.00
[root@webserver1 ~]# journalctl -xe
Nov 09 12:39:14 webserver1 ec2net[30763]: [get_meta] Querying IMDS for mac
Nov 09 12:39:14 webserver1 ec2net[30763]: [get_meta] Querying IMDS for mac
Nov 09 12:39:14 webserver1 ec2net[30763]: [get_meta] Querying IMDS for network/interfaces/macros/0a:e8:de:d0:a0:51/local-ipv4s
Nov 09 12:39:14 webserver1 ec2net[30763]: Called trap
Nov 09 12:39:14 webserver1 ec2net[30763]: No networkd reload needed
Nov 09 12:39:14 webserver1 systemd[1]: refresh-policy-routes@ens5.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://lists.freedesktop.org/mailman/listinfo/systemd-devel

The unit refresh-policy-routes@ens5.service has successfully entered the 'dead' state.
Nov 09 12:39:14 webserver1 systemd[1]: Finished refresh-policy-routes@ens5.service - Refresh policy routes for ens5.
Subject: A start job for unit refresh-policy-routes@ens5.service has finished successfully
Defined-By: systemd
```

systemctl list-units --type=service --state=failed

```
[root@webserver1 ~]# systemctl list-units --type=service --state=failed
UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.
```

## 7. Incident Management (L1 Troubleshooting)

### ◆ Issue: High CPU Usage

top -o %CPU

```
top - 12:42:28 up 2:09, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.8 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 904.8 total, 297.9 free, 222.2 used, 384.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 544.0 avail Mem
```

| PID | USER | PR | NI  | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND                     |
|-----|------|----|-----|--------|-------|-------|---|------|------|---------|-----------------------------|
| 1   | root | 20 | 0   | 172880 | 17684 | 10760 | S | 0.0  | 1.9  | 0:01.69 | systemd                     |
| 2   | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | kthreadd                    |
| 3   | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_gp                      |
| 4   | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_par_gp                  |
| 5   | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | slub_flushwq                |
| 6   | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | netns                       |
| 8   | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | kworker/0:0H-events_highpri |
| 10  | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | mm_percpu_wq                |
| 11  | root | 20 | 0   | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_tasks_kthread           |
| 12  | root | 20 | 0   | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_tasks_rude_kthread      |
| 13  | root | 20 | 0   | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | rcu_tasks_trace_kthread     |
| 14  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.14 | ksoftirqd/0                 |
| 15  | root | 20 | 0   | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.14 | rcu_preempt                 |
| 16  | root | rt | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.03 | migration/0                 |
| 18  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | cpuhp/0                     |
| 19  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | cpuhp/1                     |
| 20  | root | rt | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.06 | migration/1                 |
| 21  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.14 | ksoftirqd/1                 |
| 23  | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | kworker/1:0H-events_highpri |
| 26  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | kdevtmpfs                   |
| 27  | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | inet_frag_wq                |
| 28  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | kauditd                     |
| 29  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | khungtaskd                  |
| 30  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | oom_reaper                  |
| 32  | root | 0  | -20 | 0      | 0     | 0     | I | 0.0  | 0.0  | 0:00.00 | writeback                   |
| 33  | root | 20 | 0   | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.20 | kcompactd0                  |
| 34  | root | 39 | 19  | 0      | 0     | 0     | S | 0.0  | 0.0  | 0:00.00 | khugepaged                  |

ps aux --sort=-%cpu | head

```
[root@webserver1 ~]# ps aux --sort=-%cpu | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  1.9 172880 17684 ?        Ss   10:32   0:01 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root         2  0.0  0.0      0     0 ?        S    10:32   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        I<   10:32   0:00 [rcu_gp]
root         4  0.0  0.0      0     0 ?        I<   10:32   0:00 [rcu_par_gp]
root         5  0.0  0.0      0     0 ?        I<   10:32   0:00 [slub_flushwq]
root         6  0.0  0.0      0     0 ?        I<   10:32   0:00 [netns]
root         8  0.0  0.0      0     0 ?        I<   10:32   0:00 [kworker/0:0H-events_highpri]
root        10  0.0  0.0      0     0 ?        I<   10:32   0:00 [mm_percpu_wq]
root        11  0.0  0.0      0     0 ?        I    10:32   0:00 [rcu_tasks_kthread]
```

## ◆ Issue: Disk Full

df -h

du -sh /var/log/\* | sort -rh | head

rm -rf /var/log/old\_logs

```
[root@webserver1 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0   4.0M   0% /dev
tmpfs           453M   0   453M   0% /dev/shm
tmpfs           181M  440K  181M   1% /run
/dev/nvme0n1p1  8.0G  1.7G  6.4G  21% /
tmpfs           453M   0   453M   0% /tmp
/dev/nvme0n1p128 10M  1.3M  8.7M  13% /boot/efi
/dev/nvme1n1    10G  104M  9.9G   2% /data
tmpfs           91M   0    91M   0% /run/user/1000
[root@webserver1 ~]# du -sh /var/log/* | sort -rh | head
16M    /var/log/journal
1.7M    /var/log/dnf.librepo.log.1
292K    /var/log/dnf.log
248K    /var/log/chrony
148K    /var/log/cloud-init.log
140K    /var/log/audit
88K     /var/log/dnf.rpm.log
20K     /var/log/sa
16K     /var/log/amazon
12K     /var/log/dnf.librepo.log
[root@webserver1 ~]# rm -rf /var/log/old_logs
[root@webserver1 ~]#
```

#### ◆ Issue: Service Down

systemctl status httpd

systemctl restart httpd

journalctl -u httpd -n 100

#### ◆ Issue: Network Down

ping 8.8.8.8

curl -I https://www.google.com

ip a

systemctl restart network

---

## 8. Reporting and Documentation

### Step 15: Create Daily System Health Report

/home/adminuser/scripts/health\_report.sh

#!/bin/bash

REPORT="/home/adminuser/reports/health\_report.txt"

echo "Server Health Report - \$(date)" > \$REPORT

## Uptime Report

```
uptime >> /var/log/uptime_report.log
```

### AWS CloudWatch Metrics Check

```
aws cloudwatch get-metric-statistics \
```

```
--metric-name CPUUtilization \
```

```
--start-time 2025-11-09T00:00:00Z \
```

```
--end-time 2025-11-09T23:59:59Z \
```

```
--period 300 \
```

```
--statistics Average \
```

```
--namespace AWS/EC2 \
```

```
--dimensions Name=InstanceId,Value=i-0123456789abcdef0
```

```
Server Health Report - Sat Nov 9 13:30:01 IST 2025
```

#### Uptime:

```
13:30:01 up 10 days, 4:23, 3 users, load average: 0.15, 0.10, 0.05
```

#### Disk Usage:

| Filesystem | Size | Used | Avail | Use% | Mounted on |
|------------|------|------|-------|------|------------|
| /dev/sda1  | 100G | 40G  | 55G   | 42%  | /          |
| tmpfs      | 7.8G | 0    | 7.8G  | 0%   | /dev/shm   |

#### Memory Usage:

|      | total | used | free | shared | buff/cache | available |
|------|-------|------|------|--------|------------|-----------|
| Mem: | 15900 | 8000 | 4000 | 500    | 4900       | 7400      |

#### Top 5 Processes:

| PID  | COMMAND | %MEM | %CPU |
|------|---------|------|------|
| 1234 | java    | 12.3 | 2.1  |
| 2345 | python  | 10.2 | 1.5  |
| 3456 | mysqld  | 8.5  | 3.0  |
| 4567 | nginx   | 5.1  | 0.3  |
| 5678 | node    | 4.7  | 0.8  |